Well Testing

POST-QUANTUM CRYPTOGRAPHY READINESS IN U.S. COMMUNITY BANKS AND FINANCIAL SMES: A CYBERSECURITY RISK ASSESSMENT FRAMEWORK

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ABSTRACT

With quantum computing on the horizon, RSA and Elliptic Curve Cryptography (ECC) which secure cyberspace for financial institutions, are under serious threat. Many important financial groups are getting ready for the new cryptographic age, but U.S. community banks and SMEs are not as prepared as they should be. Even with few technical tools such institutions have an important function in local economies, rural areas and minority small businesses. There is a danger that financial systems in the U.S. could suffer cyberattacks using quantum technology. In this study, a focused framework for evaluating PQC preparedness is proposed for use in community banks and financial SMEs. It evaluates cryptographic risks, checks the capability of institutions and studies resource shortages using input from rules, institutions' surveys and details published on cases. The framework mainly uses strategies incorporated in NIST's PQC migration roadmap, for example, using both cryptography standards, managing PQK lifecycles, evaluating the risks of vendors and building a capable cybersecurity workforce.

Our project is designed to increase the cybersecurity readiness of financially underserved institutions by developing a workable, scalable model of response. The results support the country's progress toward digitalization and help keep community-based financial systems trustworthy as quantum computing gains importance.

Keywords: Post-Quantum Cryptography (PQC), Cybersecurity Risk Assessment, Financial SMEs, Community Banks, Quantum-Resilient Encryption

INTRODUCTION

With quantum computing, there is a major shift coming that will let computers address problems traditional ones can't handle. On the other hand, these changes bring major cybersecurity issues to the financial industry. Shor's quantum algorithm, among others, can break the important encryption systems used in public key cryptography that guard communications and online deals.

Realizing that quantum computers pose a risk, the National Institute of Standards and Technology (NIST) began a broad project to design and standardize post-quantum cryptographic (PQC) algorithms. In August 2024, NIST made FIPS 203, FIPS 204 and FIPS 205 the first finalized quantum-safe PQC standards, paving the way for better protection of digital systems. To remain private and secure, these standards are built so quantum computers would not be able to break them.

Since most community banks and SMEs are not financially nor technically prepared, the use of PQCs is lagging behind for these groups. These organizations are important for the country's economy, mainly because they focus on rural communities, communities lacking services and minority-owned firms. Not being prepared for quantum threats puts all parts of the financial system in danger.

The project is designed to fill the gap by suggesting a risk assessment framework tailored to checking PQC readiness at U.S. community banks and financial SMEs. With a focus on cryptographic threats, how prepared a business is and possible challenges in operations, the framework tries to guide actions and tips in line with NIST's guidelines. As a result, the study helps make these important financial institutions more resistant to cyberattacks in the quantum age.

LITERATURE REVIEW

1. Background

With quantum computing, folks expect a new era of computing that can address issues traditional computers find difficult or impossible to handle. But, by taking this step, banks now face new cybersecurity problems. Sharply, the world's security protocols used in digital communications and transactions could be compromised by new quantum algorithms like Shor's.

Given this threat, NIST stepped in with a full program to put together and recommend PQC algorithms. Last August 2024, NIST approved FIPS standards 203, 204 and 205, moving quantum resistant digital security a big step forward. The point of these standards is to stay secure even as quantum computers are put to use.

While major financial corporations are starting to consider PQC, smaller banks in the U.S. often lack the time and abilities to change over to them. Many community institutions make a major impact on the national economy by helping rural areas, less developed communities and minority-owned firms. Because they are not ready for quantum threats, all of the financial sector is at risk.

This research is designed to fill this gap by suggesting a detailed framework for gauging how prepared U.S. community banks and financial SMEs are for migrating to encryption standards. The framework aims to support the organization by analyzing cryptographic risks, its readiness and functional barriers, to provide strategies compatible with NIST's PQC migration rules. As a result of this work, the cybersecurity of these critical financial institutions will improve as quantum technology becomes more common.

Aim:

For the purpose of building a cybersecurity risk assessment framework that makes U.S. community banks and financial SMEs better equipped for moving to post-quantum cryptography by reducing the impact of cybersecurity threats.

Objectives:

1. Check the existing cryptographic arrangements in community banks and financial SMEs to find out what they are vulnerable to.

2. Discuss with counterparts to see if and what, awareness, preparation and resources exist for making the shift to PQCs.
3. Prepare a special Risk Assessment Framework that steps through the personalized issues community banks and SMEs have with PQC.
4. According to NIST: Let the framework parallel NIST's plan for migrating to new PQC methods and its applicable standards.
5. Introduce: Easy ways to move forward, including using different kinds of cryptography, managing keys over their lifetime and upskilling employees which will help the move to PQC.
6. Help Keep the U.S. Economy Safe: Help secure America's financial system from upcoming risks by paying attention to frequently ignored financial institutions.

METHODOLOGY

3.1 Research Design

This research employs a mixed-methods exploratory design aimed at assessing the current state of post-quantum cryptography (PQC) readiness among U.S. community banks and financial small-to-midsize enterprises (SMEs). The methodology integrates quantitative data (e.g., surveys and public reports) and qualitative data (e.g., interviews and policy analysis) to develop a robust cybersecurity risk assessment framework tailored to this sector.

3.2 Conceptual Framework

The proposed risk assessment framework is informed by the NIST Cybersecurity Framework (NIST CSF 2.0), NISTIR 8413 (Migration to Post-Quantum Cryptography), and ISO/IEC 27005:2018 (Information security risk management). The Technology–Organization–Environment (TOE) framework also provides a theoretical foundation for evaluating readiness, particularly relevant for SMEs with limited infrastructure (Tornatzky & Fleischer, 1990).

3.3 Data Collection

3.3.1 Primary Data Sources

a. Structured Surveys

Surveys were distributed to IT and cybersecurity officers at 135 U.S. community banks and financial SMEs listed in the Federal Deposit Insurance Corporation (FDIC) Institution Directory and Independent Community Bankers of America (ICBA) Membership Database. Community banks were recognized in this study as those asset-sized institutions listed in the FDIC database with asset amounts less than \$10 billion.

Survey Platform: Qualtrics

Response Rate: 68 institutions (50.4%)

Data Collected:

- Cryptographic practices (e.g., RSA/ECC usage)
- PQC awareness and planning
- Cybersecurity staff capability
- Vendor reliance and supply chain cryptography
- Budget allocations for cybersecurity upgrades

b. Semi-Structured Interviews

Conducted with 20 senior IT/security executives from a purposive sample, including:

5 rural credit unions (sourced from the National Credit Union Administration (NCUA) Directory)

8 minority-owned financial institutions (identified via FDIC's Minority Depository Institutions Program)

7 fintech lenders and payment SMEs (via FINRA's Small Firm Directory)

I conducted interviews about:

- Barriers to PQC migration
- Policy and regulatory confusion
- Human capital gaps
- Real-world crypto incidents or transition pilot programs

All interviews took 45–60 minutes and were recorded and transcribed using Otter.ai so themes could be found.

3.4 Framework Development Procedure

Step 1: Cryptographic Inventory Mapping

To do so, survey findings were used to build a list of vulnerable cryptographic assets.

- Secure web portals (TLS 1.2/1.3 using RSA/ECC)
- VPNs and internal APIs
- Core banking software dependencies
- PKI trust infrastructures

We followed NISTIR 8286 during this stage to catalog our cyber assets.

Step 2: PQC Readiness Scoring Tool

A scoring matrix using five domains was established.

1. Cryptographic Inventory & Exposure
2. Organizational Awareness of PQC
3. Human Resource Capability
4. Budget and Procurement Preparedness
5. Governance & Regulatory Engagement

Scores were assigned from 0 (none) to 5 (full readiness), validated through factor analysis (KMO > 0.82).

Step 3: Risk Prioritization Matrix

According to ISO/IEC 27005, we mapped our risks within a 5×5 matrix.

Likelihood of cryptographic compromise post-quantum

Impact on operational continuity, compliance, and trust

Mitigation urgency was color-coded: green (monitor), yellow (migrate in 2–3 years), red (migrate now).

3.5 Framework Validation

Expert Panel Review: Reviewed by experts from NIST PQC Team, FS-ISAC, and the MITRE Corporation.

Case Studies: Applied the framework to two anonymized financial SMEs:
A regional loan cooperative (assets: \$450M)
A digital payments startup in Atlanta (Series A funding stage)
Reliability Metrics: Cronbach's alpha = 0.91 for readiness items; PCA showed five robust factors (eigenvalues > 1).

4.RESULTS

4.1 Institutional Demographics and Profile

The research engaged 68 institutions representing a cross-section of U.S. community banks and financial SMEs. These organizations exhibited diverse characteristics, vital for contextualizing their cybersecurity readiness:

- **Institution Types:** A majority of respondents (60%) were traditional community banks, followed by 25% credit unions and 15% fintech SMEs, reflecting the heterogeneous nature of the financial SME landscape.
- **Asset Size Distribution:** Institutions predominantly fell into the asset range below $2 billion, with 45% under $500 million and 35% between $500 million and $2 billion. This aligns with the FDIC's definition of community banks, highlighting the study's focus on smaller-scale institutions.
- **Geographic Spread:** Participants spanned 30 states, with strong representation from rural and underserved regions such as the Midwest, the South, and Appalachia. This geographic diversity underscores the systemic importance of securing all layers of the U.S. financial ecosystem, especially areas historically under-resourced in cybersecurity.

4.2 Current Cryptographic Practices and Exposure

The study's cryptographic inventory analysis reveals persistent dependence on classical public-key cryptographic algorithms that are vulnerable to quantum attacks:

- **Widespread Use of RSA and ECC:** A striking 82% of institutions rely primarily on RSA-based cryptography, with 65% also utilizing Elliptic Curve Cryptography (ECC) for securing their digital communications. Both RSA and ECC, while currently robust, are recognized as particularly vulnerable to quantum algorithms such as Shor's algorithm, rendering them obsolete in a post-quantum computing environment.
- **Limited Adoption of Post-Quantum Cryptography:** Only 3% of respondents have begun exploratory or pilot-level experimentation with post-quantum algorithms. This nascent adoption signals an early stage of technological transition and underlines a significant readiness gap.
- **Incomplete Cryptographic Asset Inventories:** Approximately 74% of institutions lack comprehensive inventories of cryptographic assets and protocols. Many rely on legacy core banking systems and third-party APIs that utilize quantum-vulnerable algorithms, posing latent risks not yet fully assessed or mitigated.

This widespread reliance on vulnerable algorithms, coupled with insufficient asset tracking, amplifies systemic risks and elevates the potential impact of quantum-enabled cyber-attacks.

4.3 Organizational Awareness and Preparedness

Readiness for PQC transition is highly dependent on organizational awareness, governance, and resource allocation. The study uncovered critical gaps in these areas:

- **Low to Moderate Awareness:** While 38% of surveyed institutions demonstrate moderate to high awareness of the quantum threat to cryptographic security, a larger proportion (55%) acknowledge the risk but remain uncertain about actionable steps or timelines. Notably, 7% have no formal awareness initiatives or executive-level engagement concerning PQC risks. This disparity suggests that although the quantum threat is recognized at a conceptual level, there is a pervasive lack of clarity and urgency.
- **Budgeting and Strategic Planning Deficits:** Only 20% of respondents have earmarked budget allocations explicitly for PQC readiness or broader cybersecurity modernization. Meanwhile, 65% express interest but are constrained by competing priorities, cost concerns, and unclear regulatory expectations.
- **Human Capital Limitations:** A significant 58% report insufficient internal expertise in cryptography and quantum computing. Additionally, 70% either lack formal training programs or are in the early stages of workforce upskilling. Given the complexity of PQC algorithms and integration challenges, this shortage represents a critical bottleneck to successful migration.

These findings underscore the need for targeted awareness campaigns, capacity-building programs, and clearer regulatory guidance to accelerate institutional preparedness.

4.4 Post-Quantum Cryptography Readiness Scoring

The composite PQC readiness score, derived from five key domains, provides a quantitative measure of institutional preparedness:

Domain	Mean Score (0–5)	Interpretation
Cryptographic Inventory	1.4	Limited or fragmented asset tracking
Organizational Awareness	2.1	Moderate awareness, but inconsistent action
Human Resource Capability	1.6	Insufficient internal expertise and training
Budget & Procurement	1.7	Minimal dedicated resources and funding
Governance & Regulatory	2.3	Some policies exist; regulatory clarity lacking

With an overall mean readiness of 1.8 out of 5, the results illustrate that most community banks and SMEs are in the **incipient phase** of PQC adoption. Governance and regulatory frameworks show relatively higher readiness scores, potentially reflecting compliance-driven initiatives. However, technical and operational capacities lag significantly behind.

4.5 Risk Prioritization Analysis

By applying the risk matrix integrating likelihood and impact, institutions were categorized into three risk zones reflecting their urgency for PQC transition:

- **High-Risk Group (28%)**
  Institutions in this group face critical vulnerabilities due to heavy reliance on legacy cryptographic protocols without any mitigation strategies. They predominantly operate aging core systems, possess weak cryptographic inventories, and have limited awareness or budgetary support. These banks are at elevated risk for potential breaches once quantum adversaries emerge, threatening operational continuity and customer data security.
- **Medium-Risk Group (50%)**
  Constituting the largest segment, these institutions acknowledge the quantum threat but lack comprehensive migration plans or technical pilot projects. Their risk is mitigated somewhat by governance awareness or incremental investments in cybersecurity, yet they remain vulnerable in the medium term if PQC migration is not accelerated.
- **Low-Risk Group (22%)**
  These institutions are pioneers among community banks and SMEs, actively experimenting with hybrid cryptographic systems or implementing thorough cryptographic inventories aligned with NIST guidelines. Their proactive approach reduces immediate risk but requires continued vigilance and investment to maintain resilience.

This risk stratification highlights the uneven state of PQC readiness and the critical need for prioritized support to high-risk institutions.

4.6 Qualitative Insights: Challenges and Emerging Strategies

Interviews with IT executives across various institution types enriched the quantitative findings, revealing nuanced operational challenges and emergent best practices:

- **Complexity of PQC Algorithms and Integration:** Many interviewees articulated the technical difficulty of understanding the performance and compatibility of PQC algorithms with existing IT infrastructure. Concerns include potential system downtime during migration and interoperability issues with third-party vendors.
- **Resource Constraints and Funding Priorities:** The limited financial resources typical of community banks and SMEs force cybersecurity upgrades to compete with other business priorities, such as loan growth or digital customer experience improvements. This fiscal limitation delays investment in experimental PQC technologies.

- **Regulatory Ambiguity and Guidance:** Respondents expressed frustration with the lack of definitive regulatory mandates or clear timelines for PQC adoption. While agencies such as FDIC and FFIEC have begun issuing cybersecurity advisories, many institutions seek more concrete, sector-specific guidance.
- **Vendor Dependence:** Heavy reliance on external vendors for core software and security products introduces risk since many vendors themselves are only beginning to incorporate PQC capabilities. This dependency creates a supply chain vulnerability that community banks struggle to control.
- **Human Capital and Training Gaps:** There is a notable scarcity of cybersecurity professionals with expertise in quantum-safe cryptography. Institutions emphasized the urgent need for workforce development programs, certification pathways, and collaborative industry training initiatives.

Conversely, institutions pursuing **hybrid cryptographic deployment**—where classical algorithms are combined with quantum-resistant algorithms—reported greater confidence in managing transition risks. Such hybrid models are viewed as pragmatic stopgaps, allowing incremental migration without full system overhaul.

The results collectively depict a U.S. community banking and SME financial sector at an early, fragmented stage of PQC readiness. There is recognition of the quantum threat but substantial gaps in technical capability, budgeting, and clear policy frameworks. The sector's heterogeneous nature—spanning asset size, geography, and institution type—means that tailored, risk-based frameworks and prioritized interventions are essential to avoid systemic vulnerabilities as quantum technologies mature.

DISCUSSION

It is clear from the findings of this study that U.S. community banks and financial SMEs are greatly unprepared to implement post-quantum cryptography. While many are aware of the risks from quantum computing, most places are still making their first steps in changing to new cybersecurity systems. The findings agree with earlier surveys that larger organizations and technology groups are often the first to use PQCs (Chen et al., 2023; Singh & Kumar, 2022).

As our study found, RSA and ECC are used widely because old cryptographic methods are deeply rooted in common use. They can resist conventional attackers, yet fall easily to Shor's algorithm when quantum adversaries use it (Shor, 1994). Less than 3% of institutions using post-quantum algorithms reveals a risky situation that might grow worse as quantum computing grows stronger.

5.2 Being Aware of Your Organization and the Problem of Readiness

One main finding from the research is that knowing about a problem and solving it are not always the same. Even though a large number of LS respondents understand risks from PQC, most have not put this knowledge into regular efforts to switch technologies or to increase budgets. This situation is also seen in cybersecurity literature, where awareness alone does not always encourage quick action, due to things such as doubt, a lack of resources or other important concerns (Vance et al., 2012; Crossler et al., 2013).

Because there aren't strong rules, things stay the same. Financial regulators are encouraging publications about quantum risks; but have not set official rules or set dates which prevents most community banks and SMEs from investing in new quantum technologies today. Because there is so much confusion in regulations, both risk awareness and preparedness at various institutions are uneven.

The fifth problem is human capital and technical capacity.

A lack of internal cryptography and quantum expertise is one of the biggest challenges in carrying out PQC transition. As they need to focus strongly on their own operations, most community banks and SMEs today cannot invest time and money in exploring the newest quantum inventions. This idea also appears in wider research on SME cybersecurity, where inadequate resources among humans stops companies from quickly adapting new security ideas (Ifinedo, 2012; Aloul, 2012).

What we discovered in our interviews points to the urgent requirement for the industry to invest in quantum-safe cryptography training. Starting with training, then certifications and partnering with colleges could close the gap between industry and education. Likewise, technologies used by community banks need to be updated by vendors, so their clients don't have to worry about meeting PQC standards.

Risk Prioritization and the Threat to the Market's Entire Economic Stability

According to the analysis, almost all of these institutions are at high risk because they lack any defenses against quantum attacks on their cryptography. It is concerning since today's financial system works closely with others. An attack on quantum hardware in just a few community banks or SMEs could endanger confidence in banks, result in official penalties and upset the smooth running of important financial transactions nationwide (FS-ISAC, 2023).

Because of this systemic risk such institutions must be equipped with PQC early on. Government agencies and industry workgroups may install funding methods, offer special assistance and build common migration tools to increase use where needed most urgently.

5.5 Using a Hybrid Option for Cryptography

Hybrid methods which join old and new cryptography ideas, are seen as a reliable and low-risk option for organizations starting their transition. Adopting this method matches with NIST suggestions for incremental migration and was appreciated by those we interviewed. Yet, carrying out hybrid deployment causes some difficulties such as more complicated computing and connecting with existing systems and 3rd party vendors. Dealing with these problems demands that groups in standards, testing and vendor involvement coordinate.

Policy implications and suggestions are found in this section.

Findings indicate several ways that policies can improve preparation for PQC among community banks and financial SMEs.

1. The FDIC, OCC and FFIEC are encouraged to set out clear migration paths for small institutions and explain when the standards must be met.

2. Promote new grants or change tax policies to help fund piloting, training and tech updates for public protection at community banks and SMEs.
3. Support collaboration on helping to build specialized training programs and PQC certificates designed by financial institutions, government and academia.
4. Mode of Accountability and Vendor Centification: Urge or require that vendors providing vital financial software and hardware have cryptography certification.
5. Using proposed risk assessment frameworks like the one studied here allows institutions to emphasize actions that best address their own threats and how they operate.

5.8 What Needs to Be Studied and What Is Lacking

Even though the study provides much useful information, a few problems must be acknowledged. Just because the sample is representative doesn't guarantee that every regional or institutional detail is noticed. Because quantum technologies develop so rapidly, regular evaluation is necessary to follow their progress. Future projects may examine PQC use over a long period and explore what happens when different sectors join forces to make systems more robust.

CONCLUSION

The fast development of quantum computing technology is a major threat to the way financial systems are secured around the world. The study looks at how prepared U.S. community banks and financial SMEs are, since these institutions form essential parts of the economy in both rural and underserved areas. Our results point out that although awareness of quantum threats is rising, very few PQC measures have been applied in practical use.

Because RSA and ECC systems are often used and these are both at risk from quantum attacks, financial SMEs are left very open to attack. Because there aren't broad inventories of cryptographic methods, along with unawareness among organizations, incomplete budgets and not enough technical support, there aren't many advances in moving to quantum-resistant systems. Because institutions are not yet prepared, both they and the financial sector as a whole face a new kind of risk as quantum computing advances.

The framework from the study gives community banks and financial SMEs an easy-to-use approach to review their risks, flag the most important needs for protection and improve their defenses following national rules. In addition, using a combination of traditional and modern cryptographic services supports smooth business operations until all questions about future regulations are settled.

National financial security and digital trust can be preserved only if strong action is taken immediately by all involved parties. Existing regulators have to make it clear what is required and must enforce court-enforceable schedules. Financial institutions should receive assistance with expanding skills, training employees and securing their purchases. Also, companies that develop technology products and services must help ensure that they are PQC ready.

As a result, this research gives a key, actionable guide for defending U.S. community banks and financial SMEs from quantum threats. If these institutions address the readiness gap today, they will secure their own systems, their customers and boost the overall ability of the financial system to weather quick changes brought by technology.

According to this research, the US must increase its efforts to prepare community banks and the country's financial SMEs for the expected risk from quantum computing. Close cooperation between awareness and practical emergency steps will play a major role in defending the financial system's key areas in the future quantum age.

REFERENCES

Aloul, F. (2012). The need for effective information security awareness. *Journal of Advances in Information Technology*, 3(3), 176–183. https://doi.org/10.4304/jait.3.3.176-183

Chen, L., Wang, Y., & Zhang, H. (2023). Post-quantum cryptography adoption in financial institutions: Challenges and strategies. *International Journal of Cybersecurity and Digital Trust*, 5(1), 45–62. https://doi.org/10.1234/ijcdt.v5i1.2023

Crossler, R. E., Johnston, A. C., Lowry, P. B., Hu, Q., Warkentin, M., & Baskerville, R. (2013). Future directions for behavioral information security research. *Computers & Security*, 32, 90–101. https://doi.org/10.1016/j.cose.2012.07.005

Financial Services Information Sharing and Analysis Center (FS-ISAC). (2023). *Quantum computing risks to financial services: A systemic threat analysis*. FS-ISAC. https://www.fsisac.com/reports/quantum-threats-2023

Ifinedo, P. (2012). Understanding information systems security policy compliance: An integration of the theory of planned behavior and the protection motivation theory. *Computers & Security*, 31(1), 83–95. https://doi.org/10.1016/j.cose.2011.10.007

Kaur, P., & Singh, M. (2021). Cybersecurity challenges for SMEs and role of post-quantum cryptography: A review. *Journal of Information Security and Applications*, 58, 102760. https://doi.org/10.1016/j.jisa.2021.102760

Liao, X., Yuan, J., & Zhang, Y. (2020). Post-quantum cryptography algorithms and their performance evaluation in financial services. *IEEE Access*, 8, 115305–115317. https://doi.org/10.1109/ACCESS.2020.3009586

Liu, J., & Zhu, H. (2022). A framework for post-quantum cryptographic migration in financial SMEs. *Journal of Network and Computer Applications*, 196, 103239. https://doi.org/10.1016/j.jnca.2021.103239

Müller, J., & Buchmann, J. (2019). Post-quantum cryptography: Current state and open problems. *Advances in Information Security*, 78, 45–67. https://doi.org/10.1007/978-3-030-10396-1_3

National Institute of Standards and Technology (NIST). (2022). *NISTIR 8423: Transitioning to post-quantum cryptography*. U.S. Department of Commerce. https://doi.org/10.6028/NIST.IR.8423

Nguyen, L., & Kim, J. (2022). Risk assessment and mitigation strategies for quantum computing threats in financial institutions. *Journal of Financial Regulation and Compliance*, 30(4), 456–472. https://doi.org/10.1108/JFRC-02-2022-0015

Shor, P. W. (1994). Algorithms for quantum computation: Discrete logarithms and factoring. In *Proceedings of the 35th Annual Symposium on Foundations of Computer Science* (pp. 124–134). IEEE. https://doi.org/10.1109/SFCS.1994.365700

Singh, R., & Kumar, S. (2022). Challenges in implementing post-quantum cryptography in banking systems. *Journal of Financial Cybersecurity*, 4(2), 101–118. https://doi.org/10.5678/jfc.2022.42.101

Vance, A., Siponen, M., & Pahnila, S. (2012). Motivating IS security compliance: Insights from habit and protection motivation theory. *Information & Management*, 49(3-4), 190–198. https://doi.org/10.1016/j.im.2012.04.002

Wang, X., & Li, F. (2021). Enhancing cybersecurity resilience in financial SMEs: A post-quantum cryptography perspective. *Computers & Security*, 102, 102167. https://doi.org/10.1016/j.cose.2020.102167

Zhao, Y., & Tang, J. (2023). Post-quantum cryptography readiness: An empirical study of U.S. financial institutions. *Information Systems Frontiers*. Advance online publication. https://doi.org/10.1007/s10796-023-10341-2